

# Capacity of Additive-Noise Sticky Channels: The Bernoulli Case

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## Abstract

Sticky channels, which never destroy nor create runs, are some of the simplest types of channels with synchronization errors (such as deletions, insertions, and replications). Despite their simplicity, we know little about the capacity of the sticky channels considered in the literature so far, beyond what can be gleaned from purely numerical methods. Towards a broader and systematic exploration of sticky channels, we initiate the study of *additive-noise sticky channels*, a basic family of sticky channels that independently extend each input *run* by an amount sampled from a fixed noise distribution. The capacity of additive-noise sticky channels corresponds to the capacity per unit cost of additive-noise memoryless channels over the non-negative integers, and they capture some aspects of the loss of synchronization caused by homopolymer length estimation errors in DNA sequencing.

We focus on the simplest setting of Bernoulli additive noise with success probability  $p$ , and uncover curious behavior of the capacity beyond what numerical methods can tell us. For example, the capacity stays constant when  $p \in [1/\varphi^2, 1/2]$  with  $\varphi \approx 1.618$  the golden ratio, and behaves differently when  $p \approx 0$  vs. when  $p \approx 1$ . More generally, we determine the capacity exactly for all  $p \leq 1/2$ , and when  $1/2 < p < 1$  we give sharp analytical bounds that allow us to characterize the asymptotic behavior of the capacity as  $p \rightarrow 1$ . We also extend our analysis to the setting where input strings have bounded runlength, motivated by DNA-based data storage.

## 1 Introduction

Synchronization errors like deletions, insertions, and replications represent a fundamental challenge in information theory. Our understanding of even very simple synchronization error models remains poor, and most of the information-theoretic techniques developed over the past decades are not appropriate for dealing with such errors [Mit09, CR21].

Some of the most basic channels with synchronization errors studied so far are *sticky repeat channels* [Mit08, MTL12, ISW16, CR19]. These are channels that independently replicate each input bit a number of times dictated by some replication distribution, but always at least once. A particularly simple example is the duplication channel, which writes an input bit twice at the output with probability  $p$  and once with probability  $1 - p$ , for some duplication probability  $p \in [0, 1]$ . Because they do not introduce deletions, the capacity of every sticky repeat channel is easily seen to equal the capacity *per unit cost* (with cost function  $c(x) = x$ ) of an associated discrete memoryless channel over the positive integers, by encoding every input string as a vector of its runlengths<sup>1</sup> (and assuming that all inputs start with a 0, which does not affect the capacity) [Mit08]. The capacity

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<sup>1</sup>A *run* in a string is a maximal substring of the form  $b^\ell$  for alphabet symbol  $b$  and integer  $\ell \geq 1$ .

per unit cost with cost function  $c(x)$  of a DMC  $\text{Ch}$  with input alphabet  $\mathbb{N}_{>0} = \{1, 2, \dots\}$  is given by

$$\text{CPUC}_c(\text{Ch}) := \sup_{X: \mathbb{E}[X] < \infty} \frac{I(X; Y)}{\mathbb{E}[c(X)]},$$

where the supremum is taken over all input random variables  $X$  supported on  $\mathbb{N}_{>0}$  with finite expectation,  $Y$  denotes the output of  $\text{Ch}$  on input  $X$ , and  $I$  denotes mutual information. In this work we will only consider the special case where  $c(x) = x$ , and will refer to this simply by “capacity per unit cost” and denote it by  $\text{CPUC}(\text{Ch})$  from here onward.

For example, the capacity of the duplication channel with duplication probability  $p$  equals the capacity per unit cost of the channel that on input an integer  $n \geq 1$  outputs  $n + \text{Bin}(n, p)$ , where  $\text{Bin}(n, p)$  is a random variable following a binomial distribution with  $n$  trials and success probability  $p$ . Note that this DMC captures how the length of a given input run is changed by this noisy process.

Despite their apparent simplicity, we have thus far been unable to determine the capacity of any non-trivial sticky repeat channel. In particular, the DMCs over the positive integers resulting from the equivalence above do not have a zero cost symbol (because runlengths are strictly positive), and so Verdú’s simplified formula for computing the capacity per unit cost [Ver90] does not apply here. With a broader exploration of sticky channels in mind, we arrive at the following more basic question:

*Can we determine the capacity of some non-trivial sticky channel?*

Ideally, we would like to answer this question affirmatively for a family of sticky channels capturing interesting aspects of loss of synchronization in real-world systems.

In this work, we leverage the connection between sticky channels and DMCs over the positive integers to identify and study what we see as a basic family of sticky channels, that we call *additive-noise sticky channels*. In certain instances, we are able to exactly determine their capacity. As we discuss below, the capacity of these channels exhibits curious behavior, and they also capture aspects of the loss of synchronization stemming from sequencing errors in DNA-based data storage.

## 1.1 Additive-noise sticky channels

As discussed above, there is an equivalence between the capacity of sticky channels and the capacity per unit cost of DMCs over the positive integers. Prior work has used the equivalence to help analyze the capacity of sticky repeat channels. However, we can also turn this around and use the equivalence to find new interesting types of sticky channels by focusing on core families of DMCs, and then study their properties as a reasonable starting point towards a broader and more systematic exploration of sticky channels.

Additive-noise channels are one of the most fundamental and well-studied classes of memoryless channels. An additive-noise channel over the reals is a memoryless channel that given input  $x$  from some input alphabet  $\mathcal{X} \subseteq \mathbb{R}$  outputs  $x + Z$  where  $Z$  is a random variable distributed according to a noise distribution  $\mathcal{N}$  supported on a subset of  $\mathbb{R}$ , and “+” is the usual addition in  $\mathbb{R}$ . In this context, most prior works have focused on the case where  $\mathcal{X}$  is continuous (in particular, where  $\mathcal{X}$  is  $\mathbb{R}$  or the set of non-negative real numbers) and  $\mathcal{N}$  is a continuous distribution.<sup>2</sup> Here, we are

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<sup>2</sup>There are also well-studied discrete channels that can be framed as additive-noise channels by changing the addition operation. For example, the binary symmetric channel can be seen as an additive-noise channel if we take  $\mathcal{X} = \{0, 1\}$ ,  $\mathcal{N}$  to be Bernoulli, and “+” to mean addition modulo 2. However, here we are specifically interested in the case where “+” means addition in  $\mathbb{R}$ .

interested in the discrete counterparts of these channels, where the input alphabet  $\mathcal{X}$  is countable and the noise distribution  $\mathcal{N}$  is discrete. In particular, we focus on the case where  $\mathcal{X} = \mathbb{N}_{>0}$  and the support of  $\mathcal{N}$  is a subset of  $\mathbb{N}_{\geq 0} = \{0, 1, 2, \dots\}$ .

*Additive-noise sticky channels* are the sticky channel counterparts of additive-noise channels with positive-integer input and non-negative-integer noise, in the sense that they are the sticky channels whose capacity equals the capacity per unit cost (with cost function  $c(x) = x$ ) of such an additive-noise DMC. More precisely, we introduce the following definition.

**Definition 1.1** (Additive-noise sticky channels). *An additive-noise sticky channel is characterized by a noise distribution  $\mathcal{N}$  supported on  $\mathbb{N}_{\geq 0}$ . Given a binary input string  $x \in \{0, 1\}^n$  with  $r = r(x)$  runs of lengths  $\ell_1, \dots, \ell_r$ , the additive-noise sticky channel with noise distribution  $\mathcal{N}$ , which we denote by  $\text{ANSC}_{\mathcal{N}}$ , samples  $Z_1, \dots, Z_r$  i.i.d. according to  $\mathcal{N}$  and replaces the  $i$ -th run of  $x$  by a run of the same symbol of length*

$$L_i = \ell_i + Z_i$$

*for each  $i \in \{1, 2, \dots, r\}$ .*

We are interested in studying the (coding) capacity of  $\text{ANSC}_{\mathcal{N}}$ . This capacity equals the capacity per unit cost of an additive-noise memoryless channel, as described more explicitly in the following lemma. Its proof is analogous to that of Mitzenmacher [Mit08, Theorem 2.1]. For completeness, we provide a proof in [Appendix A](#).

**Lemma 1.2.** *Let  $\mathcal{N}$  be a distribution supported on  $\mathbb{N}_{\geq 0}$ . Then, the coding capacity of  $\text{ANSC}_{\mathcal{N}}$  equals the capacity per unit cost of the memoryless channel  $\text{ANMC}_{\mathcal{N}}$  with input alphabet  $\mathbb{N}_{>0}$  which on input  $n$  outputs  $n + Z$  with  $Z$  distributed according to  $\mathcal{N}$ .*

## 1.2 Additive-noise sticky channels and homopolymer length estimation in DNA-based data storage

DNA-based data storage is an emerging technology boasting much higher storage density and durability than currently used systems. In those systems, data (encoded in a 4-letter alphabet) is synthesized into DNA strands, with each letter mapped to a *nucleotide*. Then, sequencing technologies are used to recover stored data from these strands. DNA sequencing is an error-prone process that introduces various types of errors, from substitutions and erasures to errors that cause loss of synchronization, such as deletions, insertions, and replications.

It is well known that long runs of nucleotides (called *homopolymers*, corresponding to long runs in the stored string) are particularly problematic for widely used sequencing technologies, such as Oxford Nanopore Technologies and Illumina [EZ17, YGM17, WGGH23]. In nanopore-based sequencing, stored symbols are inferred from an electrical current signal generated as a DNA strand sequentially passes through a biological pore. The measured signal is continuous and, at any given time, depends on a short window of consecutive nucleotides within the pore. The window shifts by one position stochastically, and these shifts are detected by changes in the current signal. Long homopolymers generate an almost constant signal, and their length must be inferred from the duration of this constant signal. Since shifting is not a deterministic process, there is a chance of under- or over-estimating the length of the homopolymer (equivalently, the length of the corresponding run in the stored string), leading to loss of synchronization [RKD18]. Similar limitations arise in Illumina sequencing, where long homopolymers impair the accurate resolution of repeated incorporations due to signal saturation and phasing effects [SN21]. Overall, homopolymers increase the likelihood of synchronization errors, with error rates typically increasing with the run length [YGM17, WGGH23, CR25, CMRS26].

We may see additive-noise sticky channels as a toy model that captures some important aspects of the noisy process of homopolymer length estimation, making some simplifying assumptions compared to practice. First, the model only considers replications (i.e., overestimation of runlengths). Second, the model assumes that the number of errors within a run does not depend on its length nor on the number of errors in surrounding runs. We believe that these simplifications are a reasonable starting point for studying noisy homopolymer length estimation, as it still allows us to capture non-trivial aspects of the loss of synchronization resulting from that process. Some other recent works have considered non-sticky synchronization errors and runlength-dependent error probabilities in the context of homopolymer length estimation, but they either focus on the incomparable setting of adversarial errors [CMRS26] or only provide loose capacity lower bounds [CR25]. As we shall see below, we are able to carry out a much sharper study of the properties of our channels (naturally, because our model is simpler).

### 1.3 An overview of our contributions

We initiate the study of the capacity of additive-noise sticky channels by focusing on Bernoulli noise. That is, we take  $\mathcal{N}$  to be a Bernoulli distribution with some success probability  $p \in [0, 1]$ . We denote the corresponding channel by  $\text{ANSC}_{\text{Ber}(p)}$  and its coding capacity by  $C(p)$ . By [Lemma 1.2](#),  $C(p) = \text{CPUC}(\text{ANMC}_{\text{Ber}(p)})$ , where we recall that  $\text{ANMC}_{\text{Ber}(p)}$  is the additive-noise memoryless channel that on input  $n \in \mathbb{N}_{>0}$  outputs  $n + Z$  with  $Z$  a Bernoulli random variable with success probability  $p$ .

As with all sticky channels studied so far, it is not hard to numerically approximate  $C(p)$  for any given  $p$  with great accuracy (we discuss this in [Section 5](#)). However, such an approach provides limited insights about the channel. Our goal, then, is to develop an understanding of the capacity that is out of reach of numerical methods, either by determining  $C(p)$  exactly or by deriving asymptotic results.

For the  $\text{ANSC}_{\text{Ber}(p)}$  there is a simple baseline lower bound on  $C(p)$  obtained by considering only input strings  $x \in \{0, 1\}^n$  with runs of odd length. It is clear that such a code is zero-error over the  $\text{ANSC}_{\text{Ber}(p)}$ , since an input run of length  $\ell$  is mapped to a run of length either  $\ell$  or  $\ell + 1$  at the output. As we show in [Section 3](#), this naive *runlength-constrained* coding strategy yields the lower bound

$$C(p) \geq \log \varphi \approx 0.694$$

valid for any  $p \in [0, 1]$ , where  $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$  is the golden ratio and  $\log$  is the base-2 logarithm. Some questions arise:

1. Is naive runlength-constrained coding ever optimal?
2. How much better can we do than runlength-constrained coding?

**Optimality of naive runlength-constrained coding.** We settle the first question by characterizing the values of  $p$  for which naive runlength-constrained coding is optimal.

**Theorem 1.3.** *We have  $C(p) \geq \log \varphi$  for all  $p \in [0, 1]$ , and  $C(p) = \log \varphi$  if and only if  $p \in \left[\frac{1}{\varphi^2} \approx 0.382, \frac{1}{2}\right]$ .*

For more details, see [Section 3](#). In particular, this means that naive runlength-constrained coding is optimal even when the noise is somewhat biased towards 0, but immediately becomes sub-optimal when the noise is biased towards 1.

**The  $p \leq 1/2$  setting.** With the first question answered, we move towards understanding the second question. We go beyond [Theorem 1.3](#) and determine  $C(p)$  (along with a capacity-achieving input distribution) for all  $p \leq 1/2$ .

**Theorem 1.4.** *When  $p \leq 1/2$  we have*

$$C(p) = \begin{cases} \log \varphi, & \text{if } p \in \left[\frac{1}{\varphi^2}, \frac{1}{2}\right], \\ \log x_p, & \text{if } p \in \left[0, \frac{1}{\varphi^2}\right), \end{cases}$$

where  $x_p$  is the unique solution to the equation  $x_p - 2^{-h(p)}x_p^p - 1 = 0$  with  $h(p) = -p \log p - (1-p) \log(1-p)$  the binary entropy function.

For more details, see [Sections 4.1](#) and [4.2](#).

**The  $p > 1/2$  setting.** In this regime we are not able to exactly determine  $C(p)$ , but we derive sharp bounds that, in particular, allow us to pinpoint the asymptotic behavior of  $C(p)$ .

**Theorem 1.5.** *Fix  $p \in (1/2, 1]$ . For  $\beta \in \mathbb{R}$ , denote by  $\lambda(p, \beta)$  the unique solution to the equation*

$$\sum_{n=1}^{\infty} 2^{-\lambda(p, \beta)(n-p) - h(p) - \left(-\frac{1-p}{p}\right)^{n-1} \beta} = 1.$$

*Then,  $C(p) \leq \inf_{\beta \in \mathbb{R}} \lambda(p, \beta)$ .*

For more details, see [Sections 4.1](#) and [4.3](#). Based on a comparison with numerical approximations of  $C(p)$  in [Section 5](#), we conjecture that the upper bound in [Theorem 1.5](#) is actually the capacity. We formalize this in [Conjecture 5.1](#) and leave it as an interesting direction for future work.

We use [Theorem 1.5](#) to better understand the asymptotic behavior of  $C(p)$  when  $p \rightarrow 1$  and when  $p \rightarrow \frac{1}{2}$  from above:

- In the regime where  $p = 1 - \varepsilon$  with  $\varepsilon \rightarrow 0$ , we can show using [Theorem 1.5](#) and a geometric input distribution that

$$C(1 - \varepsilon) = 1 - (1 + o(1)) \frac{h(\varepsilon)}{4},$$

where  $o(1) \rightarrow 0$  as  $\varepsilon \rightarrow 0$ . For details, see [Section 4.3.1](#). It is interesting to compare  $C(1 - \varepsilon)$  and  $C(\varepsilon)$  when  $\varepsilon \rightarrow 0$ , as perhaps one would expect the capacity to behave similarly in both of these regimes. However, it is not hard to see that  $C(\varepsilon) = 1 - (1 + o(1)) \frac{h(\varepsilon)}{2}$ , and so  $C(1 - \varepsilon) > C(\varepsilon)$  for all sufficiently small  $\varepsilon$ .

- In the regime where  $p = \frac{1}{2} + \varepsilon$  with  $\varepsilon \rightarrow 0$  from above we also obtain two-sided bounds on  $C(p)$  that, while not being sharp, give us concrete estimates on the strict improvement over naive runlength-constrained coding. More precisely, we show that

$$\log \varphi \cdot \left(1 + \log \varphi \cdot \frac{4(3 - \varphi) \ln 2}{5\sqrt{5}(\varphi - 1)} \cdot \varepsilon^2\right) - O(\varepsilon^3) \leq C\left(\frac{1}{2} + \varepsilon\right) \leq \log \varphi \cdot \left(1 + \frac{2\varepsilon}{2 + \varphi}\right) + O(\varepsilon^2).$$

The lower bound is obtained by considering an appropriate convex combination of geometric-like input distributions. For details, see [Section 4.3.2](#).

**Capacity under bounded runlength.** Motivated by the connection to DNA data storage, where data has to be encoded into codewords with bounded runlengths due to various system constraints, we extend our study of the  $\text{ANSC}_{\text{Ber}(p)}$  to the setting where inputs  $x \in \{0, 1\}^n$  only have runs of length at most some integer  $\tau \geq 1$ . When the noise is Bernoulli with success probability  $p$ , we denote the capacity in this setting by  $C(p, \tau)$ . As before, our goal is to obtain insights about  $C(p, \tau)$  beyond what can be extracted from purely numerical methods.

When  $\tau$  is an odd integer, we prove analogs of [Theorems 1.3](#) and [1.4](#) for  $C(p, \tau)$ , in particular characterizing the region where naive runlength-constrained coding is optimal. For  $p \leq \frac{1}{2}$ , we show that  $C(p, \tau)$  approaches  $C(p)$  exponentially fast in  $\tau$ . When  $\tau$  is even, we find that naive runlength-constrained coding is never optimal. Thus, we are not able to determine the exact capacity for all values of  $\tau$  when  $p$  belongs to the interval given in [Theorem 1.3](#). Moreover, for  $p$  in the interior of that interval, we prove that the capacity-achieving distribution also does not have full support, as long as  $\tau$  is even and large enough. Details can be found in [Section 6](#).

## 1.4 More related work

As discussed above, studying the capacity of additive-noise sticky channels is equivalent to studying the capacity per unit cost of additive-noise memoryless channels with discrete alphabet  $\mathbb{N}_{>0}$  and noise supported on  $\mathbb{N}_{\geq 0}$ . In turn, the capacity per unit cost is closely connected to the average power-constrained capacity, where we focus on inputs  $X$  such that  $\mathbb{E}[X] \leq \mu$  for some  $\mu$ . The average power-constrained capacity of additive-noise memoryless channels over the non-negative integers has appeared before in the study of discrete-time queuing channels [[BA98](#), [PG03](#)]. There, they only consider the case where the additive noise is geometric.

## 1.5 Directions for future research

Our work leaves open several directions for future exploration. We highlight a few of them here:

- Settle [Conjecture 5.1](#), i.e., whether the bound given in [Theorem 1.5](#) is actually the capacity of the  $\text{ANSC}_{\text{Ber}(p)}$  for  $p > 1/2$ .
- Carry out an analogous study for a more general class of additive-noise sticky channels.
- Add more components to the model to more accurately capture noisy homopolymer estimation in DNA-based data storage systems.

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# 2 Preliminaries

## 2.1 Notation

We denote random variables by uppercase roman letters such as  $X$ ,  $Y$ , and  $Z$ . Sets are usually denoted by calligraphic uppercase letters such as  $\mathcal{S}$  and  $\mathcal{T}$ . We write  $\mathbb{N}_{>0}$  for the set of positive



integers and  $\mathbb{N}_{\geq 0}$  for the set of non-negative integers. We also write  $[n]$  for the set  $\{1, 2, \dots, n\}$ . We denote the base-2 logarithm by  $\log$  and the natural logarithm by  $\ln$ .

We will deal solely with discrete random variables. Let  $X$  be a random variable supported on  $\mathcal{X} \subseteq \mathbb{R}$  and some  $x \in \mathcal{X}$ . We write  $P_X$  for the probability mass function of  $X$  and  $P_X(x)$  for the probability that  $X$  takes on value  $x$ . We denote the expectation of  $X$  by  $\mathbb{E}[X]$ . For two arbitrarily correlated random variables  $X$  and  $Y$  we will use the notation  $P_{Y_x}$  as shorthand for  $P_{Y|X=x}$ .

## 2.2 Characterization of the capacity per unit cost of memoryless channels

Recall [Lemma 1.2](#) relating the capacity of additive-noise sticky channels to the capacity per unit cost of additive-noise memoryless channels over the non-negative integers. Motivated by this, the following framework due to Abdel-Ghaffar [\[Abd93\]](#) will be useful to us.

**Lemma 2.1** ([\[Abd93\]](#)). *Let  $\text{Ch}$  be a memoryless channel with countable input and output alphabets  $\mathcal{X}, \mathcal{Y} \subseteq \mathbb{N}_{>0}$ , respectively, and  $c : \mathcal{X} \rightarrow \mathbb{R}_{>0}$  a positive cost function. Denote by  $Y_x$  the output random variable of  $\text{Ch}$  on input  $x \in \mathcal{X}$ . Then, we have*

$$\text{CPUC}_c(\text{Ch}) \leq \sup_{x \in \mathcal{X}} \frac{D_{\text{KL}}(P_{Y_x} \| Q)}{c(x)}$$

for any distribution  $Q$  supported on  $\mathcal{Y}$ , where  $D_{\text{KL}}(P \| Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$  is the Kullback-Leibler divergence between discrete distributions  $P$  and  $Q$ .

Furthermore, if there exists an input distribution  $P_X$  such that  $Q$  is its corresponding output distribution with respect to  $\text{Ch}$  and

$$\frac{D_{\text{KL}}(P_{Y_x} \| Q)}{c(x)} \leq \lambda$$

for all  $x \in \mathcal{X}$  with equality when  $P_X(x) > 0$ , then  $\text{CPUC}_c(\text{Ch}) = \lambda$  and  $P_X$  is capacity-achieving.

The proof of [Lemma 2.1](#) presented by Abdel-Ghaffar [\[Abd93\]](#) implicitly assumes that the input and output alphabets are finite. However, it is not hard to see that it extends to countably infinite alphabets too.<sup>3</sup>

When the input and output alphabets are finite, the sufficient conditions for optimality above are also necessary.

**Lemma 2.2** ([\[Abd93\]](#)). *Recall the setting of [Lemma 2.1](#). If  $\mathcal{X}$  and  $\mathcal{Y}$  are finite and  $P_X$  with corresponding output distribution  $P_Y$  achieves the capacity per unit cost of  $\text{Ch}$  with cost function  $c$ , then*

$$\frac{D_{\text{KL}}(P_{Y_x} \| P_Y)}{c(x)} \leq \text{CPUC}_c(\text{Ch})$$

for all  $x \in \mathcal{X}$  with equality for all  $x$  such that  $P_X(x) > 0$ .

<sup>3</sup>For example, we can invoke a general result of Lapidoth and Moser [\[LM03, Theorem 5.1\]](#), which states that in the setting of [Lemma 2.1](#) (and well beyond it) we have  $I(X; Y) \leq \sum_{x \in \mathcal{X}} P_X(x) D_{\text{KL}}(P_{Y_x} \| Q)$  for any input distribution  $P_X$  and any distribution  $Q$  over  $\mathcal{Y}$ , with  $Y$  the corresponding channel output. Then, if  $D_{\text{KL}}(P_{Y_x} \| Q) \leq \lambda c(x)$  for all  $x$ , we get that

$$\frac{I(X; Y)}{\mathbb{E}[c(X)]} \leq \frac{\sum_{x \in \mathcal{X}} P_X(x) D_{\text{KL}}(P_{Y_x} \| Q)}{\mathbb{E}[c(X)]} \leq \frac{\lambda \sum_{x \in \mathcal{X}} P_X(x) c(x)}{\mathbb{E}[c(X)]} = \lambda.$$

### 3 When is naive runlength-constrained coding optimal?

In this section we prove [Theorem 1.3](#), thus characterizing the optimality region of the the naive runlength-constrained coding strategy of taking only strings with runs of odd length as codewords. We restate the theorem here.

**Theorem 1.3.** *We have  $C(p) \geq \log \varphi$  for all  $p \in [0, 1]$ , and  $C(p) = \log \varphi$  if and only if  $p \in \left[\frac{1}{\varphi^2} \approx 0.382, \frac{1}{2}\right]$ .*

*Proof.* We consider the input distribution

$$P_X(n) = \begin{cases} \varphi^{-n}, & \text{if } n \text{ is odd,} \\ 0, & \text{otherwise.} \end{cases}$$

First, note that

$$C(p) = \text{CPUC}(\text{ANMC}_{\text{Ber}(p)}) \geq \frac{I(X; Y)}{\mathbb{E}[X]} = \frac{H(X)}{\mathbb{E}[X]} = \log \varphi.$$

The second equality uses the fact that  $H(X|Y) = 0$ , since  $X$  is supported on the odd integers and  $Y \in \{X, X+1\}$  with probability 1. This proves the first part of the theorem.

To see the second part, we use [Lemma 2.1](#) to determine when  $X$  is capacity-achieving. Note that

$$\sum_{n=1}^{\infty} P_X(n) = \sum_{k=1}^{\infty} \varphi^{-(2k-1)} = 1,$$

and so  $P_X$  is a valid probability mass function. With this choice of  $X$ , the corresponding output  $Y$  has probability mass function

$$P_Y(n) = \begin{cases} (1-p)P_X(n), & \text{if } n \text{ is odd,} \\ p \cdot P_X(n-1), & \text{otherwise.} \end{cases}$$

We now compute  $D_{\text{KL}}(P_{Y_n} \| P_Y)$  for an arbitrary integer  $n > 0$ , where we recall that  $P_{Y_n}$  denotes the channel output distribution on input  $n$ . For odd  $n$ , we get

$$\begin{aligned} D_{\text{KL}}(P_{Y_n} \| P_Y) &= (1-p) \log \left( \frac{1}{P_Y(n)} \right) + p \log \left( \frac{1}{P_Y(n+1)} \right) - h(p) \\ &= (1-p) \log \left( \frac{1}{(1-p)P_X(n)} \right) + p \log \left( \frac{1}{p \cdot P_X(n)} \right) - h(p) \\ &= \log \left( \frac{1}{P_X(n)} \right) \\ &= n \cdot \log \varphi. \end{aligned}$$

For even  $n$ , we get

$$\begin{aligned} D_{\text{KL}}(P_{Y_n} \| P_Y) &= (1-p) \log \left( \frac{1}{P_Y(n)} \right) + p \log \left( \frac{1}{P_Y(n+1)} \right) - h(p) \\ &= (1-p) \log \left( \frac{1}{p \cdot P_X(n-1)} \right) + p \log \left( \frac{1}{(1-p)P_X(n+1)} \right) - h(p) \\ &= (1-p)(n-1) \log \varphi + p(n+1) \log \varphi - (1-p) \log p - p \log(1-p) - h(p) \end{aligned}$$



$$= n \cdot \log \varphi + (2p - 1) \log \varphi - (1 - p) \log p - p \log(1 - p) - h(p).$$

Define  $\Delta(p) = (2p - 1) \log \varphi - (1 - p) \log p - p \log(1 - p) - h(p)$ . We get that  $D_{\text{KL}}(P_{Y_n} \| P_Y) \leq \lambda n$  if and only if  $\Delta(p) \leq 0$ , in which case we conclude that  $X$  is capacity-achieving. By routine algebraic manipulation, we conclude that this holds if and only if

$$\frac{1}{\varphi^2} \leq p \leq \frac{1}{2}.$$

Consequently, for  $p$  in this interval we get via [Lemma 2.1](#) that

$$C(p) = \text{CPUC}(\text{ANMC}_{\text{Ber}(p)}) = \log \varphi. \quad \square$$

## 4 A capacity upper bound for general $p$

In this section we derive an upper bound on  $C(p)$  that holds for arbitrary  $p$ . For  $p < 1/\varphi^2$  we show that this upper bound actually *is* the capacity, and so, together with [Theorem 1.3](#), we determine  $C(p)$  exactly for  $p \in [0, 1/2]$ , leading to [Theorem 1.4](#). For  $p > 1/2$  we leave it as a conjecture that our upper bound is the capacity, supported by numerical evidence. Nevertheless, we are able to use our upper bound in that setting to characterize the asymptotic behavior of  $C(p)$  as  $p \rightarrow 1$ .

### 4.1 Setting things up

Our strategy is to “guess” that the capacity-achieving output  $Y$  must have full support over  $\mathbb{N}_{>0}$ . More precisely, with [Lemma 2.1](#) in mind, we will characterize the distributions  $Q$  that satisfy

$$D_{\text{KL}}(P_{Y_n} \| Q) = \lambda n, \quad \text{for all } n \in \mathbb{N}_{>0}, \quad (1)$$

for some  $\lambda$ . By that lemma, we conclude that all such  $\lambda$ ’s provide an upper bound on  $C(p) = \text{CPUC}(\text{ANMC}_{\text{Ber}(p)})$ .

Without loss of generality, we write

$$Q(n) = 2^{-\lambda(n-p)-h(p)-g(n)}$$

for some  $\lambda$  and function  $g$ . Note that not all choices of  $\lambda$  and  $g$  are allowed, since it must be the case that  $\sum_{n=1}^{\infty} Q(n) = 1$ . Writing  $Q$  in this way is useful because

$$\begin{aligned} D_{\text{KL}}(P_{Y_n} \| Q) &= (1 - p) \log(1/Q(n)) + p \log(1/Q(n+1)) - h(p) \\ &= \lambda n + (1 - p)g(n) + pg(n+1). \end{aligned}$$

Therefore, [Equation \(1\)](#) is equivalent to  $g$  satisfying

$$(1 - p)g(n) + pg(n+1) = 0$$

for all  $n \geq 1$ , which is equivalent to  $g(n+1) = -\frac{1-p}{p} \cdot g(n)$  for all  $n \geq 1$ . By induction, we conclude that for [Equation \(1\)](#) to hold it must be that

$$g(n) = \left(-\frac{1-p}{p}\right)^{n-1} g(1)$$

for all  $n \geq 1$ .

We summarize this discussion in the following theorem, writing  $\beta = g(1)$ .

**Theorem 4.1.** Fix  $p \in [0, 1]$  and  $\lambda, \beta \in \mathbb{R}$ . Suppose that

$$Q(n) = 2^{-\lambda(n-p)-h(p)-\left(-\frac{1-p}{p}\right)^{n-1}\beta}$$

satisfies  $\sum_{n=1}^{\infty} Q(n) = 1$ . Then,  $C(p) \leq \lambda$ . Furthermore, if there exists an input distribution  $P_X$  that yields  $Q$  as the output distribution over the  $\text{ANMC}_{\text{Ber}(p)}$ , then  $C(p) = \lambda$ .

## 4.2 The $p \leq 1/2$ setting

We will now use [Theorem 4.1](#) to determine  $C(p)$  for all  $p \leq 1/2$ , leading to [Theorem 1.4](#), which we restate here.

**Theorem 1.4.** When  $p \leq 1/2$  we have

$$C(p) = \begin{cases} \log \varphi, & \text{if } p \in \left[\frac{1}{\varphi^2}, \frac{1}{2}\right], \\ \log x_p, & \text{if } p \in [0, \frac{1}{\varphi^2}), \end{cases}$$

where  $x_p$  is the unique solution to the equation  $x_p - 2^{-h(p)}x_p^p - 1 = 0$  with  $h(p) = -p \log p - (1-p) \log(1-p)$  the binary entropy function.

*Proof.* The part about  $p \geq 1/\varphi^2$  was already established in [Theorem 1.3](#), so we focus on the setting where  $p < 1/\varphi^2$ .

Note that when  $p < 1/2$  the quantity  $\left(\frac{1-p}{p}\right)^{n-1}$  grows exponentially with  $n$ . Therefore, in this setting, the only choice of  $\beta$  that makes  $Q$  in [Theorem 4.1](#) a valid probability mass function is  $\beta = 0$ , and so we focus on

$$Q(n) = 2^{-\lambda(n-p)-h(p)}. \quad (2)$$

Since it must be the case that  $\sum_{n=1}^{\infty} Q(n) = 1$ , we conclude that  $\lambda$  must be the (unique) solution to

$$\sum_{n=1}^{\infty} Q(n) = \frac{2^{\lambda \cdot p - h(p)}}{2^{\lambda} - 1} = 1, \quad (3)$$

in which case  $C(p) \leq \lambda$ .

By [Theorem 4.1](#), to prove that  $C(p) = \lambda$  it suffices to show that  $Q$  with  $\beta = 0$  and  $\lambda$  the unique solution to [Equation \(3\)](#) can be realized as an output distribution of the  $\text{ANMC}_{\text{Ber}(p)}$ . This holds if there exists  $X$  such that we can write  $Q$  as the distribution of  $X + Z$  with  $Z$  an independent  $\text{Ber}(p)$  random variable. We can use this to obtain a candidate generating function  $G_X$  for  $P_X$ . This candidate generating function is not guaranteed to be a *probability generating function*, in the sense that its terms need not all be non-negative. We then show that this is indeed the case, and so conclude that the desired  $X$  exists.

Let  $G$  denote the probability generating function of  $Q$ . For  $Z \sim \text{Ber}(p)$  independent of  $X$  and  $w \in \mathbb{C}$  we have

$$G(w) = \mathbb{E}[w^Y] = \mathbb{E}[w^{X+Z}] = \mathbb{E}[w^X] \cdot \mathbb{E}[w^Z] = G_X(w)(1-p+pw).$$

Therefore, we must have

$$G_X(w) = \frac{G(w)}{1-p+pw}$$

in the disk  $\{w \in \mathbb{C} : |w| < 1\}$  (recall that  $p \leq 1/2$ ).

Recalling Equation (2), it is not hard to see that  $G$  admits the closed-form

$$G(w) = \frac{2^{\lambda p - h(p)} w}{2^\lambda - w},$$

and so

$$G_X(w) = \frac{2^{\lambda p - h(p)} w}{(1 - p + pw)(2^\lambda - w)}. \quad (4)$$

Note that  $G_X(w)$  is analytic in the disk  $|w| < 1$ . To see that  $P_X$  is a valid probability distribution it is enough to show that the coefficients of Taylor expansion of  $G_X(z)$  around  $w = 0$  are all non-negative and that the order-0 coefficient is 0, since we know that  $\sum_{n \geq 1} P_X(n) = G_X(1) = \frac{G(1)}{1-p+p} = 1$ . To simplify the exposition, we ignore the  $2^{\lambda p - h(p)}$  multiplicative factor in Equation (4) as it does not depend on  $n$  and is always positive. In other words, we have to show that the coefficients of the Taylor expansion of

$$\frac{w}{(1 - p + pw)(2^\lambda - w)}$$

around  $w = 0$  are always positive. It is not hard to see by induction that the order- $n$  coefficient of the Taylor expansion of  $\frac{w}{(1-p+pw)(2^\lambda-w)}$  around  $w = 0$  is

$$\alpha_n = \sum_{i=0}^{n-1} \frac{(-p)^i}{2^{\lambda(n-i)}(1-p)^{i+1}} = \frac{1}{2^{\lambda n}(1-p)} \sum_{i=0}^{n-1} \left(-\frac{p2^\lambda}{1-p}\right)^i = \frac{1 - \left(-\frac{p2^\lambda}{1-p}\right)^n}{2^{\lambda n}(1-p+p2^\lambda)}.$$

Note that  $\alpha_0 = 0$  and, for  $n \geq 1$ , we have  $\alpha_n \geq 0$  (and hence  $P_X(n) \geq 0$ ) if and only if

$$1 \geq \left(-\frac{p2^\lambda}{1-p}\right)^n.$$

This is clearly true when  $n$  is odd, because the right-hand side is always non-positive in that case. When  $n$  is even, this is equivalent to

$$0 \leq \lambda \leq \log\left(\frac{1-p}{p}\right).$$

To see this, recall that  $\lambda$  is the unique solution to Equation (3). Since  $\lambda \mapsto \sum_{n \geq 1} Q(n)$  is a decreasing function of  $\lambda > 0$ , to verify Equation (3) it suffices to check that replacing  $\lambda$  by  $\log\left(\frac{1-p}{p}\right)$  makes the left-hand side at most 1, i.e., that

$$\frac{2^{\log\left(\frac{1-p}{p}\right)p - h(p)}}{2^{\log\left(\frac{1-p}{p}\right)} - 1} = \frac{p(1-p)}{1-2p} \leq 1.$$

This holds if  $0 \leq p \leq 1/\varphi^2$ . Finally, to get the desired statement in the theorem define  $x_p = 2^\lambda$  and rewrite Equation (3) in terms of  $x_p$ .  $\square$

### 4.3 The $p > 1/2$ setting

Note that for any fixed  $p \in [1/2, 1]$  the term  $\left(\frac{1-p}{p}\right)^{n-1}$  decreases exponentially fast to 0 as  $n \rightarrow \infty$ . Therefore, we get that for every  $p \in [1/2, 1]$  and  $\beta \in \mathbb{R}$  there is a (unique)  $\lambda$  such that

$$\sum_{n=1}^{\infty} Q(n) = \sum_{n=1}^{\infty} 2^{-\lambda(n-p)-h(p)-\left(-\frac{1-p}{p}\right)^{n-1}\beta} = 1.$$

Therefore, invoking [Theorem 4.1](#) with  $p \in [1/2, 1]$  directly yields [Theorem 1.5](#), which we restate here.

**Theorem 1.5.** *Fix  $p \in (1/2, 1]$ . For  $\beta \in \mathbb{R}$ , denote by  $\lambda(p, \beta)$  the unique solution to the equation*

$$\sum_{n=1}^{\infty} 2^{-\lambda(p, \beta)(n-p)-h(p)-\left(-\frac{1-p}{p}\right)^{n-1}\beta} = 1.$$

*Then,  $C(p) \leq \inf_{\beta \in \mathbb{R}} \lambda(p, \beta)$ .*

#### 4.3.1 Asymptotic behavior of $C(p)$ as $p \rightarrow 1$

In this section we use [Theorem 1.5](#) to determine the asymptotic behavior of  $C(p)$  as  $p \rightarrow 1$  up to low-order terms, as described in the following theorem.

**Theorem 4.2.** *We have*

$$C(1 - \varepsilon) = 1 - (1 + o(1)) \frac{h(\varepsilon)}{4},$$

*where  $o(1) \rightarrow 0$  as  $\varepsilon \rightarrow 0$ .*

*Proof.* We prove this theorem in two stages. First, we show that for all  $\varepsilon > 0$  small enough we have  $C(1 - \varepsilon) \leq 1 - \frac{h(\varepsilon)}{4}$ . Second, we analyze the rate achieved by a geometric input distribution, and show that it is  $1 - (1 + o(1)) \frac{h(\varepsilon)}{4}$ . Together, these results yield the desired theorem statement.

To get the upper bound on  $C(p)$  for  $p = 1 - \varepsilon$  large we instantiate [Theorem 1.5](#) with  $\beta = \frac{ph(p)}{1-p}$ . consider the function  $f(\lambda, p)$  defined as

$$f(\lambda, p) = \sum_{n=1}^{\infty} 2^{-\lambda(n-p)-h(p)-g(n)} = \sum_{n=1}^{\infty} 2^{-\lambda(n-p)-h(p)-\left(-\frac{1-p}{p}\right)^{n-1} \frac{ph(p)}{1-p}}.$$

From [Theorem 1.5](#), we know that for each  $p \in (1/2, 1)$  the unique  $\lambda_p > 0$  such that  $f(\lambda_p, p) = 1$  is an upper bound on  $C(p)$ . Furthermore, with  $p$  fixed the map  $\lambda \mapsto f(\lambda, p)$  is decreasing in  $\lambda$ . Our goal is to show that

$$\lambda_p \leq 1 - \frac{h(p)}{4} \tag{5}$$

for all sufficiently large  $p < 1$ , which implies the same upper bound on  $C(p)$  via the discussion above. We will derive [Equation \(5\)](#) in two steps:

1. We upper bound  $f(\lambda, p)$  by a more tractable function  $\tilde{f}(\lambda, p)$ .
2. We show that  $\tilde{f}(1 - h(p)/4, p) \leq 1$  for all sufficiently large  $p < 1$ . This implies that  $f(1 - h(p)/4, p) \leq 1$  for all sufficiently large  $p < 1$  too. Then, since  $\lambda \mapsto f(\lambda, p)$  is decreasing in  $\lambda$ , we can conclude that  $\lambda_p \leq 1 - h(p)/4$  for all sufficiently large  $p < 1$ .

We obtain  $\tilde{f}$  via the following claim.

**Claim 4.3.** *For all  $\lambda > 0$  and  $p \in (1/2, 1)$  we have that*

$$f(\lambda, p) \leq 2^{-\lambda(1-p)-\frac{h(p)}{1-p}} + 2^{-\lambda(2-p)} + 2^{-\lambda(3-p)-\frac{h(p)}{p}} + \frac{2^{-\lambda(3-p)-h(p)+\left(\frac{1-p}{p}\right)^2 h(p)}}{2^\lambda - 1} =: \tilde{f}(\lambda, p).$$

*Proof.* We may write

$$f(\lambda, p) = 2^{-\lambda(1-p) - \frac{h(p)}{1-p}} + 2^{-\lambda(2-p)} + 2^{-\lambda(3-p) - \frac{h(p)}{p}} + \sum_{n=4}^{\infty} 2^{-\lambda(n-p) - h(p) - \left(-\frac{1-p}{p}\right)^{n-1} \frac{ph(p)}{1-p}}. \quad (6)$$

Then, note that  $\left(\frac{1-p}{p}\right)^{n-1}$  is decreasing in  $n$  for fixed  $p > 1/2$ . Therefore,

$$2^{-\lambda(n-p) - h(p) - \left(-\frac{1-p}{p}\right)^{n-1} \frac{ph(p)}{1-p}} \leq 2^{-\lambda(n-p) - h(p) + \left(\frac{1-p}{p}\right)^2 h(p)}$$

for all  $n \geq 4$ , and so

$$\sum_{n=4}^{\infty} 2^{-\lambda(n-p) - h(p) - \left(-\frac{1-p}{p}\right)^{n-1} \frac{ph(p)}{1-p}} \leq \sum_{n=4}^{\infty} 2^{-\lambda(n-p) - h(p) + \left(\frac{1-p}{p}\right)^2 h(p)} = \frac{2^{-\lambda(3-p) - h(p) + \left(\frac{1-p}{p}\right)^2 h(p)}}{2^\lambda - 1}. \quad (7)$$

Combining [Equations \(6\) and \(7\)](#) yields the desired upper bound.  $\square$

Define  $\gamma(p) = \tilde{f}(1 - h(p)/4, p)$ . Then, it is not hard to check that  $\gamma'(1) = \ln 2 - 1/e$ , and so

$$\gamma(p) = 1 - (\ln 2 - 1/e)(1-p) + O((1-p)^2).$$

Since  $\ln 2 > 1/e$ , it follows that there exists  $p_0 < 1$  such that  $\gamma(p) = \tilde{f}(1 - h(p)/4, p) < 1$  for all  $p \in (p_0, 1)$ . As discussed above, this implies that

$$C(p) \leq \lambda_p \leq 1 - h(p)/4 \quad (8)$$

for all  $p \in (p_0, 1)$ .

We now turn to lower bounding  $C(p)$  for large  $p$ . When  $p = 1$  the capacity-achieving input distribution  $X$  is geometric with  $P_X(n) = 2^{-n}$ . Therefore, we should expect that this input distribution gives a fairly sharp lower bound when  $p \rightarrow 1$ . Note that  $\mathbb{E}[X] = 2$ , and so the achievable rate is

$$\frac{I(X; Y)}{\mathbb{E}[X]} = \frac{H(Y) - H(Y|X)}{2} = \frac{H(Y) - h(p)}{2}.$$

where  $Y$  is the output satisfying  $P_Y(1) = (1-p)P_X(1) = \frac{1-p}{2}$  and, for  $n \geq 2$ ,

$$P_Y(n) = p \cdot P_X(n-1) + (1-p)P_X(n) = p2^{-n+1} + (1-p)2^{-n} = 2^{-n}(1+p).$$

Therefore,

$$\begin{aligned} H(Y) &= \frac{1-p}{2} \log\left(\frac{2}{1-p}\right) + \sum_{n \geq 2} P_Y(n) \log(1/P_Y(n)) \\ &= \frac{1-p}{2} \log\left(\frac{2}{1-p}\right) + (1+p) \sum_{n \geq 2} 2^{-n} (n - \log(1+p)) \\ &= \frac{1-p}{2} \log\left(\frac{2}{1-p}\right) + (1+p) \left(\frac{3}{2} - \frac{\log(1+p)}{2}\right) \\ &= 1 + p + h\left(\frac{1-p}{2}\right). \end{aligned}$$

Consequently, we have the lower bound

$$C(p) \geq \frac{1 + p + h\left(\frac{1-p}{2}\right) - h(p)}{2} = 1 - (1 + o(1)) \frac{h(p)}{4}. \quad (9)$$

The theorem statement follows by combining [Equations \(8\) and \(9\)](#) and noting that for  $p = 1 - \varepsilon$  we have  $h(p) = h(\varepsilon)$ .  $\square$

A plot of the closed-form upper and lower bounds obtained in the proof of [Theorem 4.2](#) are given in [Figure 2a](#) together with a numerical approximation of  $C(p)$ .

### 4.3.2 Asymptotic behavior of $C(p)$ as $p \rightarrow 1/2$

In this section we study the asymptotic behavior of  $C(p)$  in the regime where  $p > 1/2$  and  $p \rightarrow 1/2$ . First we present an upper bound on the capacity.

**Theorem 4.4.** *For  $\varepsilon \rightarrow 0$  from above, we have*

$$C\left(\frac{1}{2} + \varepsilon\right) \leq \log \varphi \cdot \left(1 + \frac{2\varepsilon}{2 + \varphi}\right) + O(\varepsilon^2).$$

*Proof.* Let  $p = \frac{1}{2} + \varepsilon$ . To get the upper bound on  $C(\frac{1}{2} + \varepsilon)$ , we instantiate [Theorem 1.5](#) with a careful choice of  $\beta > 0$ . First, we upper bound  $Q(n)$  by a more tractable expression. Note that for all odd  $n \geq 1$  we have

$$\begin{aligned} Q(n) &= 2^{-\lambda(n-p)-h(p)-\left(\frac{1-p}{p}\right)^{n-1}\beta} \\ &\leq 2^{-\lambda(n-p)-h(p)-(1-4\varepsilon)^{n-1}\beta} \\ &\leq 2^{-\lambda(n-p)-h(p)-\beta+4(n-1)\varepsilon\beta}, \end{aligned}$$

while for all even  $n \geq 2$  we have

$$\begin{aligned} Q(n) &= 2^{-\lambda(n-p)-h(p)+\left(\frac{1-p}{p}\right)^{n-1}\beta} \\ &\leq 2^{-\lambda(n-p)-h(p)-\beta+\frac{\beta}{p}}. \end{aligned}$$

Therefore

$$\begin{aligned} 1 &= \sum_{n=1}^{\infty} Q(n) \leq \sum_{n=1}^{\infty} 2^{-\lambda(2n-1-p)-h(p)-\beta+4(2n-2)\varepsilon\beta} + \sum_{n=1}^{\infty} 2^{-\lambda(2n-p)-h(p)-\beta+\frac{\beta}{p}} \\ &= 2^{\lambda p - h(p) - \beta} \left( \frac{2^{\lambda - 8\varepsilon\beta}}{2^{2\lambda - 8\varepsilon\beta} - 1} + \frac{2^{\frac{\beta}{p}}}{2^{2\lambda} - 1} \right). \end{aligned} \quad (10)$$

We will now take

$$\beta = 1 + \frac{\log \varphi}{2} - h(p). \quad (11)$$

To motivate this choice of  $\beta$ , note that for  $\varepsilon = 0$  the upper bound in [Equation \(10\)](#) is minimized by the choice in [Equation \(11\)](#). Now, denote

$$\lambda(p) = \log \varphi + \Delta(p). \quad (12)$$

for some function  $\Delta$  satisfying  $\Delta(1/2) = 0$ , and denote

$$f(p) = 2^{\lambda p - h(p) - \beta} \left( \frac{2^{\lambda - 8\varepsilon\beta}}{2^{2\lambda - 8\varepsilon\beta} - 1} + \frac{2^{\frac{\beta}{p}}}{2^{2\lambda} - 1} \right).$$

Note that if  $f(p) \leq 1$ , then  $\lambda \leq \lambda(p)$  for that choice of  $p$  because the right-hand side of Equation (10) is decreasing with  $\lambda$  and greater than 1. Since we are interested in the behavior around  $p = 1/2$ , we focus on the corresponding Taylor expansion

$$f(p) = 1 + \left(p - \frac{1}{2}\right) \left( (1 - 2\varphi) \Delta' \left(\frac{1}{2}\right) + 2(\varphi - 1) \log \varphi \right) \ln 2 + O \left( \left(p - \frac{1}{2}\right)^2 \right).$$

This means that  $f(p) \leq 1$  for  $p$  sufficiently close to  $1/2$  provided that

$$\Delta' \left(\frac{1}{2}\right) \geq \frac{2(\varphi - 1)}{2\varphi - 1} \log \varphi = \frac{2}{\varphi + 2} \log \varphi.$$

Now, note that

$$\Delta(p) = \left( \frac{2}{\varphi + 2} \log \varphi + \delta \right) \left( p - \frac{1}{2} \right) \quad (13)$$

satisfies  $\Delta(1/2) = 0$  and  $\Delta'(1/2) \geq 2 \log \varphi / (\varphi + 2)$  for any  $\delta > 0$ . Injecting Equation (13) in (12), we conclude that

$$\lambda \leq \log \varphi \cdot \left( 1 + \left( \frac{2}{\varphi + 2} + \delta \right) \left( p - \frac{1}{2} \right) \right) + O \left( \left( p - \frac{1}{2} \right)^2 \right),$$

for any  $\delta > 0$ , provided that  $p$  is sufficiently close to  $1/2$  (depending on  $\varepsilon$ ).  $\square$

To complement the asymptotic upper bound above, we present an asymptotic lower bound on the capacity in the regime  $p \rightarrow 1/2$ .

**Theorem 4.5.** *For  $\varepsilon > 0$  we have*

$$C \left( \frac{1}{2} + \varepsilon \right) \geq \log \varphi \cdot \left( 1 + \log \varphi \cdot \frac{4(3 - \varphi) \ln 2}{5\sqrt{5}(\varphi - 1)} \varepsilon^2 \right) - O(\varepsilon^3).$$

*Proof.* Fix  $0 < \varepsilon \leq 1/2$  and define  $p = 1/2 + \varepsilon$ . Let  $0 < \gamma \leq 2$ , to be chosen later. We consider a mixture  $P_X$  with parameter  $\gamma\varepsilon$  of two distributions  $P_{X_1}$  and  $P_{X_2}$ , given by

$$P_X = (1 - \gamma\varepsilon)P_{X_1} + \gamma\varepsilon P_{X_2}.$$

We choose the first distribution of the mixture to be the runlength-constrained coding input distribution

$$P_{X_1}(n) = \begin{cases} \varphi^{-n}, & \text{if } n \text{ is odd,} \\ 0, & \text{otherwise,} \end{cases}$$

which is capacity achieving for  $1/\varphi^2 \leq p \leq 1/2$  (see Section 3) and induces the output distribution

$$P_{Y_1}(n) = \begin{cases} (1 - p)\varphi^{-n}, & \text{if } n \text{ is odd,} \\ p\varphi^{-(n-1)}, & \text{otherwise.} \end{cases}$$



The second input distribution of the mixture is

$$P_{X_2}(n) = \begin{cases} 0, & \text{if } n \text{ is odd,} \\ 2^{-\frac{n}{2}} & \text{otherwise,} \end{cases}$$

and induces the output distribution

$$P_{Y_2}(n) = \begin{cases} 0, & \text{if } n = 1, \\ p2^{-\frac{n-1}{2}}, & \text{if } n \text{ is odd, } n \geq 3, \\ (1-p)2^{-\frac{n}{2}}, & \text{otherwise.} \end{cases}$$

Recall that  $p = \frac{1}{2} + \varepsilon$ , a Taylor expansion of the mutual information around  $\varepsilon = 0$  gives

$$\begin{aligned} I(X; Y) &= \frac{(\gamma\varepsilon)^2}{2\ln 2} + (1 - \gamma\varepsilon)I(X_1; Y_1) - \gamma\varepsilon h(p) - \gamma\varepsilon \sum_{n \geq 1} P_{Y_2}(n) \log P_{Y_1}(n) \\ &\quad - \frac{1}{2\ln 2} \frac{(\gamma\varepsilon)^2}{1 - \gamma\varepsilon} \sum_{n \geq 1} \frac{P_{Y_2}(n)^2}{P_{Y_1}(n)} - O(\varepsilon^3). \end{aligned} \quad (14)$$

We now expand each term in Equation (14) asymptotically. First, we have

$$h(p) = 1 - O(\varepsilon^2).$$

Second, we have

$$\begin{aligned} \sum_{n \geq 1} P_{Y_2}(n) \log P_{Y_1}(n) &= - \sum_{n \geq 1} P_{Y_2}(n) - \log \varphi \sum_{n \geq 1} P_{Y_2}(n)n + \log \varphi \sum_{\text{even } n, n \geq 2} P_{Y_2}(n) \\ &\quad + 2\varepsilon \sum_{\text{even } n, n \geq 2} P_{Y_2}(n) - 2\varepsilon \sum_{\text{odd } n, n \geq 1} P_{Y_2}(n) - O(\varepsilon^2) \\ &= -1 - 4\log \varphi - 2\log \varphi \cdot \varepsilon - O(\varepsilon^2). \end{aligned} \quad (15)$$

Third, we have

$$\sum_{n \geq 1} \frac{P_{Y_2}(n)^2}{P_{Y_1}(n)} = \frac{3\varphi + 1}{2(3 - \varphi)} + O(\varepsilon).$$

The reasoning behind this particular mixture choice is the following. In order to maximize the right-hand side of Equation (14), we want to maximize the expectation of  $Y_2$  while minimizing  $\sum_{\text{even } n, n \geq 2} P_{Y_2}(n)$  in Equation (15). Since  $\mathbb{P}[Y_2 \text{ is even}] = (1-p)\mathbb{P}[X_2 \text{ is even}] + p \cdot \mathbb{P}[X_2 \text{ is odd}] = 1 - p + (2p - 1)\mathbb{P}[X_2 \text{ is odd}]$  and  $p > \frac{1}{2}$ , we choose a distribution  $P_{X_2}$  with weight only on even numbers.

Recalling that the RLL achievable rate per unit cost is  $\log \varphi$  (see Section 3), we obtain from Equation (14) that

$$\begin{aligned} I(X; Y) &= (1 - \gamma\varepsilon) \log \varphi \cdot \mathbb{E}[X_1] - \gamma\varepsilon - \gamma\varepsilon(-1 - 4\log \varphi - 2\log \varphi \cdot \varepsilon) + \frac{(\gamma\varepsilon)^2}{2\ln 2} \left( 1 - \frac{3\varphi + 1}{2(3 - \varphi)} \right) + O(\varepsilon^3) \\ &= (1 - \gamma\varepsilon)\sqrt{5} \log \varphi + 4\gamma \log \varphi \cdot \varepsilon + (\gamma\varepsilon)^2 \left( \frac{5(1 - \varphi)}{4(3 - \varphi)\ln 2} + \frac{2\log \varphi}{\gamma} \right) + O(\varepsilon^3), \end{aligned}$$

where

$$\mathbb{E}[X_1] = \sqrt{5}.$$

Then, noting that  $\mathbb{E}[X] = \sqrt{5}\left(1 + \gamma\varepsilon\frac{4-\sqrt{5}}{\sqrt{5}}\right)$ , the achievable rate per unit cost of the mixture is

$$\frac{I(X;Y)}{\mathbb{E}[X]} = \log \varphi + \frac{(\gamma\varepsilon)^2}{\sqrt{5}} \left( \frac{5(1-\varphi)}{4(3-\varphi)\ln 2} + 2\frac{\log \varphi}{\gamma} \right) - O(\varepsilon^3). \quad (16)$$

It remains to optimize the choice of  $\gamma$ . The optimal choice is  $\gamma^* = \frac{4(3-\varphi)\ln 2 \cdot \log \varphi}{5(\varphi-1)}$ , which satisfies  $\gamma^* \leq 2$  ensuring  $\gamma\varepsilon \leq 1$  for  $0 \leq \varepsilon \leq 1/2$ . Plugging this choice into Equation (16) leads to the desired statement.  $\square$

A plot of the closed-form upper and lower bounds obtained in Theorem 4.4 and Theorem 4.5 together with a numerical approximation of  $C(p)$  is given in Figure 2b.

## 5 Numerical approximation of $C(p)$ and a conjecture

While Theorem 1.4 determines  $C(p)$  for all  $p \leq 1/2$ , Theorem 1.5 provides only an upper bound on  $C(p)$  for  $p > 1/2$ . In this section we discuss a numerical approximation of  $C(p)$ , which in particular provides evidence that the upper bound from Theorem 1.5 is the capacity. More precisely, we formulate the following conjecture.

**Conjecture 5.1.** Fix  $p \in (1/2, 1]$ . For  $\beta \in \mathbb{R}$ , denote by  $\lambda(p, \beta)$  the unique solution to the equation

$$\sum_{n=1}^{\infty} 2^{-\lambda(p, \beta)(n-p) - h(p) - \left(-\frac{1-p}{p}\right)^{n-1} \beta} = 1.$$

Then,  $C(p) = \inf_{\beta \in \mathbb{R}} \lambda(p, \beta)$ .

Towards numerically approximating  $C(p)$ , we leverage Lemma 1.2 and employ a variant of the well-known Blahut–Arimoto algorithm (BAA) [Bla72, Ari72], specifically the version developed by Jimbo and Kunisawa [JK79]. Recall that the  $\text{ANMC}_{\text{Ber}(p)}$  takes as input a natural number  $n \in \mathbb{N}_{>0}$  and outputs  $n + Z$  with  $Z$  following a Bernoulli distribution with success probability  $p$ . In what follows, we numerically approximate this capacity by studying a finite-input, finite-output truncation of this channel.

Namely, we will numerically approximate the capacity of the channel  $\text{ANMC}_{\text{Ber}(p), \tau}$  which gets as input only  $n \in [\tau]$  and outputs  $n + \text{Ber}(p)$ . This channel and its capacity is discussed in more detail in Section 6. For the purpose of this section, we observe that this channel is “equivalent” to the channel  $\text{ANSC}_{\text{Ber}(p), \tau}$  which gets as input  $x \in \{0, 1\}^n$  having runs of length at most  $\tau$  and applies  $\text{ANSC}_{\text{Ber}(p)}$  to the input. Formally, if we denote by  $C(p, t)$  the capacity of  $\text{ANSC}_{\text{Ber}(p), \tau}$ , then, by Lemma 1.2 (whose proof can be found in the appendix), we have that

$$C(p, \tau) = \text{CPUC}(\text{ANMC}_{\text{Ber}(p), \tau}) = \sup_{X: \mathbb{E}[X] < \infty} \frac{I(X; Y)}{\mathbb{E}[X]},$$

where the supremum is taken over all input random variables supported on  $[\tau]$ , with finite expectation, and  $Y$  is the output of  $\text{ANMC}_{\text{Ber}(p), \tau}$  on input  $X$ .

In general, when a channel is a DMC with finite input and output alphabets, one can apply the Blahut–Arimoto algorithm [Bla72, Ari72] to numerically approximate its capacity. In our setting,

the channel  $\mathcal{C}_{p,N}^{\text{BAS}}$  is indeed a DMC with finite input and output alphabets; however, our focus is on the capacity per unit cost. Therefore, we employ the standard modification of the BAA proposed by Jimbo and Kunisawa [JK79] to approximate the capacity per unit cost of the channel.

In both the BAA and the algorithm of Jimbo and Kunisawa, the algorithm gets as input the channel's transition law  $P(y | x)$ , which specifies, for every  $x \in \mathcal{X}$  and  $y \in \mathcal{Y}$ , the probability that the channel outputs  $y \in \mathcal{Y}$  when  $x \in \mathcal{X}$  is transmitted. The algorithm is initialized with some input distribution  $X^{(0)}$  supported on the input alphabet, and at each iteration this distribution is updated. It is known that the resulting sequence of distributions  $X^{(0)}, X^{(1)}, X^{(2)}, \dots$  converges to an input distribution that attains the capacity of the corresponding DMC (see [JK79, Theorem 1] for capacity per unit cost version). In our implementation, we also give as input a stopping criteria which will govern the error term to capacity. We summarize it in the following theorem.

**Theorem 5.2** ([JK79, Theorem 1, Corollary 2]). *Let  $a > 0$ ,  $p \in [0, 1]$ , and  $\tau \in \mathbb{N}$ . There exists an iterative algorithm that outputs a distribution  $P_X$  supported on  $[\tau]$  such that*

$$\frac{I(X; Y)}{\mathbb{E}[X]} \leq C(p, \tau) \leq \frac{I(X; Y)}{\mathbb{E}[X]} + a .$$

Moreover, the number of iterations is  $O(1/a)$ .

Our next goal is to show that  $C(p, \tau)$  closely approximates  $C(p)$ , particularly when  $\tau$  is large. To establish this, we will rely on the following simple lemma and a claim.

We will make use of a standard result stating that among all random variables on  $\mathbb{N}$  with a given finite mean, the one with maximal entropy is the geometric random variable having that same mean [Con04, Theorem 5.8].

**Lemma 5.3.** *Let  $X$  be a random variable sampled from a distribution supported on  $\{1, 2, 3, \dots\}$  with finite mean. Then,  $H(X) \leq \log(e \cdot \mathbb{E}[X])$ .*

*Proof.* By [Con04, Theorem 5.8],  $H(X) \leq H(Y)$  where  $Y \sim \text{Geom}(1/\mathbb{E}[X])$ , and we have

$$H(Y) = -\log(\mathbb{E}[X]) - (\mathbb{E}[X] - 1) \log\left(\frac{\mathbb{E}[X]}{\mathbb{E}[X] - 1}\right) \leq \log(e \cdot \mathbb{E}[X]) . \quad \square$$

Next, we show that any distribution which achieves capacity on the  $\text{ANMC}_{\text{Ber}(p)}$  must have “small” expectation.

**Claim 5.4.** *Let  $P_X$  be a distribution which achieves capacity on  $\text{ANMC}_{\text{Ber}(p)}$ . It holds that  $\mathbb{E}[X] < 6$ .*

*Proof.* Assume that  $\mathbb{E}[X] \geq 6$ . Since  $P_X$  achieves capacity,

$$C(p) = \frac{I(X; Y)}{\mathbb{E}[X]} \leq \frac{H(X)}{\mathbb{E}[X]} \leq \frac{\log(e\mathbb{E}[X])}{\mathbb{E}[X]} \leq \frac{\log(6e)}{6} \leq 0.672 ,$$

where the second inequality is due to Lemma 5.3 and the third inequality follows since  $\log(ex)/x$  is decreasing in the range  $x \geq 1$ . This contradicts the fact that the capacity of this channel is at least  $\log \varphi \geq 0.697$  for every  $p \in [0, 1]$ .  $\square$

Using these two elementary bounds and the simple claim above, we now show that  $C(p, \tau)$  provides a fairly tight approximation to  $C(p)$ . In particular, the approximation error is on the order of  $O(\frac{\log \tau}{\tau})$ .

**Theorem 5.5.** *For every  $\tau \geq 1$ , we have that*

$$C(p, \tau) \leq C(p) \leq C(p, \tau) + \frac{7 \log(e(\tau + 1)) - 11}{\tau + 1}.$$

*Proof.* The lower bound is immediate, since any distribution supported on  $[\tau]$  is also a valid distribution on  $\mathbb{N}$ . Consequently, the supremum taken over all finite-mean distributions supported on  $\mathbb{N}$  is at least as large as the supremum taken over finite-mean distributions supported on  $[\tau]$ .

For the upper bound, let  $P_X$  be a distribution which achieves capacity on  $\text{ANMC}_{\text{Ber}(p)}$  and let  $X \sim P_X$  be a random variable. Denote  $A = \mathbf{1}\{X > \tau\}$ , i.e.,  $A$  is the event that  $X$  is larger than  $\tau$  and denote  $\alpha = \Pr[A]$ . Since  $A$  is determined by  $X$ , we have that  $I(X; Y) = I(A, X; Y)$ . Now, by the chain rule, we have

$$\begin{aligned} I(A, X; Y) &= I(A; Y) + I(X; Y|A) \\ &= I(A; Y) + \alpha I(X; Y|A = 1) + (1 - \alpha) I(X; Y|A = 0). \end{aligned}$$

Thus, we have

$$C(p) \leq \frac{I(A; Y) + \alpha I(X; Y|A = 1) + (1 - \alpha) I(X; Y|A = 0)}{\mathbb{E}[X]}. \quad (17)$$

We will upper bound each term separately. For the first term we have

$$\frac{I(A; Y)}{\mathbb{E}[X]} \leq h(\alpha) \leq h\left(\frac{6}{\tau + 1}\right) \leq \frac{6(\log(e(\tau + 1)) - 2)}{\tau + 1}. \quad (18)$$

Here, the first inequality follows since  $I(A; Y) \leq H(A) = h(\alpha)$  and we have that  $\mathbb{E}[X] \geq 1$ . The second inequality uses the Markov inequality which gives that  $\alpha = \Pr[X \geq \tau + 1] \leq \frac{\mathbb{E}[X]}{\tau + 1} \leq \frac{6}{\tau + 1}$  where the last inequality is due to [Claim 5.4](#). The last inequality is by the well-known upper bound  $h(x) \leq x \log(e/x)$ .

Now, the second term can be upper bounded as follows

$$\begin{aligned} \frac{\alpha \cdot I(X; Y|A = 1)}{\mathbb{E}[X]} &\leq \frac{H(Y|A = 1)}{\mathbb{E}[X|A = 1]} \\ &\leq \frac{H(X|A = 1) + h(p)}{\mathbb{E}[X|A = 1]} \\ &\leq \frac{\log(e(\mathbb{E}[X|A = 1])) + h(p)}{\mathbb{E}[X|A = 1]} \\ &\leq \frac{\log(e(\tau + 1)) + h(p)}{\tau + 1}. \end{aligned} \quad (19)$$

The first inequality follows since

$$\frac{\alpha}{\mathbb{E}[X]} = \frac{\alpha}{(1 - \alpha) \cdot \mathbb{E}[X|A = 0] + \alpha \cdot \mathbb{E}[X|A = 1]} \leq \frac{1}{\mathbb{E}[X|A = 1]}.$$

The second inequality follows by recalling that  $Y = X + \text{Ber}(p)$  and that  $H(X + \text{Ber}(p)) \leq H(X) + H(\text{Ber}(p))$ . The third inequality is due to [Lemma 5.3](#) and the fourth inequality follows since  $\log(ex)/x$  is a decreasing function for  $x \geq 1$  and  $\mathbb{E}[X|A = 1] \geq \tau + 1$ .

Finally, the third term can be upper bounded by

$$\frac{(1 - \alpha) \cdot I(X; Y|A = 0)}{\mathbb{E}[X]} \leq \frac{(1 - \alpha) \mathbb{E}[X|A = 0] \cdot C(p, \tau)}{\mathbb{E}[X]} \leq C(p, \tau). \quad (20)$$

The first inequality follows since the distribution of  $(X|A = 0)$  is supported on  $[\tau]$  and thus  $I(X; Y|A = 0)/\mathbb{E}[X|A = 0] \leq C(p, \tau)$ . The second inequality follows since

$$\frac{(1 - \alpha) \cdot \mathbb{E}[X|A = 0]}{\mathbb{E}[X]} = \frac{(1 - \alpha) \sum_{n \in [\tau]} n \cdot \frac{P_X(n)}{(1 - \alpha)}}{\mathbb{E}[X]} = \frac{\sum_{n \in [\tau]} n \cdot P_X(n)}{\mathbb{E}[X]} \leq 1.$$

Plugging [Equations \(18\) to \(20\)](#) into [Equation \(17\)](#) yields the claimed proposition.  $\square$

Using [Proposition 5.5](#), we can apply the Jimbo–Kunisawa algorithm ([Theorem 5.2](#)) to obtain numerical estimates of  $C(p)$  and then compare these values with those predicted by our [Conjecture 5.1](#). Specifically, we run the algorithm from [Theorem 5.2](#) with parameters  $\tau = 2^{16}$  and  $a = 10^{-3}$ . This produces a lower bound on  $C(p, 2^{16})$  that also serves as a tight approximation of the capacity, with at most  $10^{-3}$  additive error. A plot of this lower bound is given in [Figure 1](#).

Moreover, when combined with [Proposition 5.5](#), this yields a numerical approximation of  $C(p)$  whose additive error is at most  $3 \cdot 10^{-3}$ . For several values of  $p$ , [Table 1](#) compares these capacity approximations with our conjectured values. Furthermore, [Figure 2](#) presents a comparison between the approximated capacity and the asymptotic upper and lower bounds obtained in previous sections, in the regimes where  $p \approx 1/2$  and  $p \approx 1$ .

| $p$  | Approximation | UB ( <a href="#">Theorem 1.5</a> ) | $p$  | Approximation | UB ( <a href="#">Theorem 1.5</a> ) |
|------|---------------|------------------------------------|------|---------------|------------------------------------|
| 0.55 | 0.6970746     | 0.6970751                          | 0.80 | 0.7863129     | 0.7863129                          |
| 0.60 | 0.7051158     | 0.7051163                          | 0.85 | 0.8208797     | 0.8208797                          |
| 0.65 | 0.7179734     | 0.7179744                          | 0.90 | 0.8635515     | 0.8635517                          |
| 0.70 | 0.7355886     | 0.7355888                          | 0.95 | 0.9178513     | 0.9178514                          |
| 0.75 | 0.7581924     | 0.7581924                          | 0.99 | 0.9774431     | 0.9774431                          |

Table 1: A comparison between our numerical estimates of the capacity  $C(p)$  and the values obtained from the upper bound in [Theorem 1.5](#), which we conjecture ([Conjecture 5.1](#)) to equal the true capacity, shows consistency to at least five decimal digits.

## 6 The Bernoulli additive-noise sticky channel with bounded run-lengths

We now consider the Bernoulli additive-noise sticky channel with error probability  $p \leq \frac{1}{2}$ , and input runlength bounded from above by  $\tau < \infty$  (this is equivalent to considering the corresponding additive-noise memoryless channel with finite input alphabet  $[\tau]$ ). We are able to determine the capacity  $C(p, \tau)$  for all  $p \leq \frac{1}{2}$  when  $\tau$  is odd. When  $\tau$  is even, we determine the exact capacity for values of  $p$  contained in a subinterval of  $[0, \frac{1}{2}]$ , and for the remaining region of  $[0, \frac{1}{2}]$  we obtain upper and lower bounds which approach the exact capacity exponentially fast as  $\tau$  increases.

### 6.1 When is naive runlength-constrained coding optimal?

We find that the naive input distribution over strings with runs of odd length is optimal when  $\tau$  is odd, for a region of  $p$  which approaches the plateau region  $[\frac{1}{\varphi^2}, \frac{1}{2}]$  for the infinite input alphabet setting found in [Theorem 1.3](#). Moreover, both the capacity and the plateau region for  $p$  with input alphabet  $[\tau]$  approach their infinite-alphabet analogues exponentially fast in  $\tau$ . When  $\tau$  is even, the

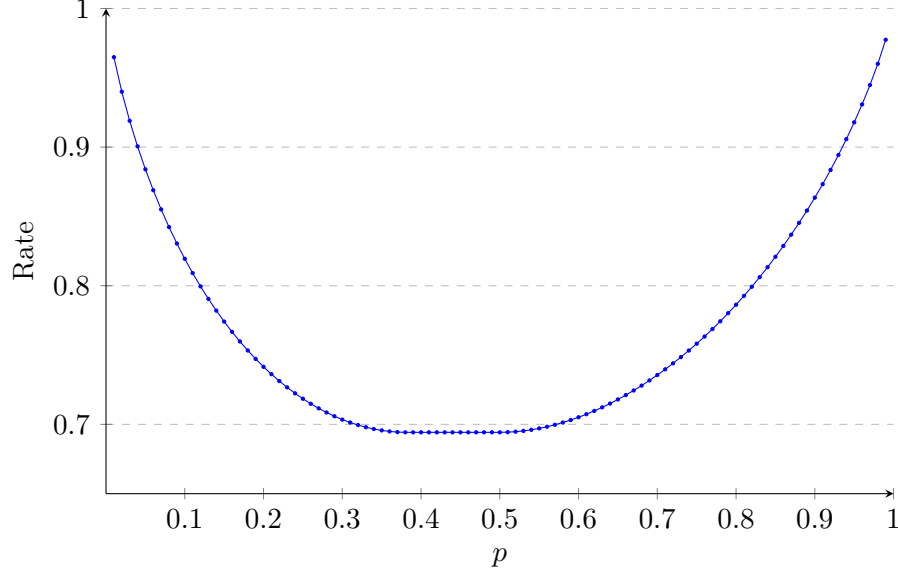
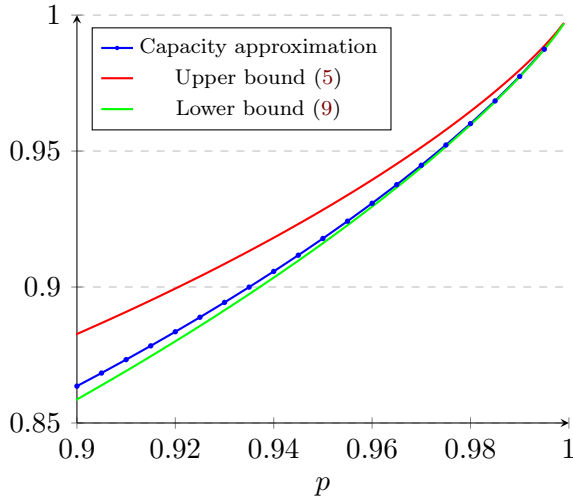
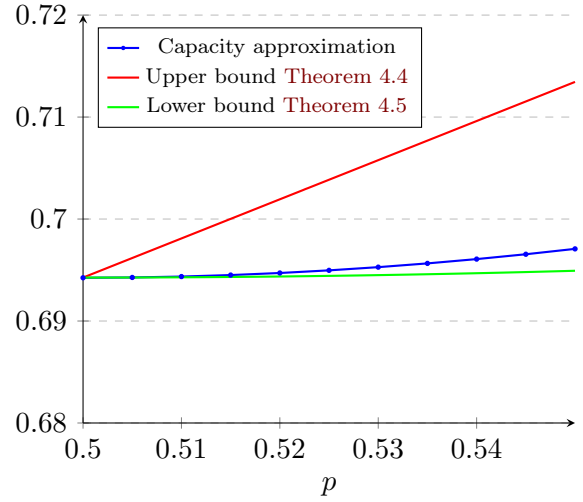


Figure 1: A numerical lower bound on the capacity of  $C(p, \tau)$  for  $\tau = 2^{16}$ , computed using the Jimbo–Kunisawa algorithm (Theorem 5.2) with threshold  $a = 0.001$ . As discussed in Section 5, these points approximate  $C(p)$  with an additive error of less than 0.003.



(a) Our asymptotically optimal closed-form upper and lower bounds (derived in Section 4.3) compared to the numerical approximation of the capacity in the large  $p$  regime.



(b) Our asymptotic closed-form upper and lower bounds compared to the numerical approximation of the capacity in the  $p \approx 1/2$  regime. When plotting these bounds, we omitted the lower order terms.

Figure 2: Comparison of our closed-form bounds with numerical capacity approximations in the regimes  $p \approx 1$  and  $\approx 1/2$ .

naive coding strategy is not optimal. While we do not obtain the exact capacity for this case, we remark that the trivial bounds

$$C(p, \tau - 1) \leq C(p, \tau) \leq C(p, \tau + 1)$$

hold for all  $p, \tau$ .

**Theorem 6.1.** *Fix an odd integer  $\tau > 1$  and let  $\lambda_\tau \in (0, 1]$  be the unique solution to*

$$2^{2\lambda} - 2^\lambda - 1 + 2^{-\lambda\tau} = 0.$$

*Then,  $C(p, \tau) \geq \lambda_\tau$  with equality if and only if  $p \in \left[\frac{1}{2^{\lambda_\tau+1}}, \frac{1}{2}\right]$ . In particular,*

$$C(p, \tau) > \log \varphi - 1.3^{-\tau}$$

*for  $p \in \left[\frac{1}{\varphi^2}(1 + 1.3^{-\tau}), \frac{1}{2}\right]$ .*

*Proof.* The proof is analogous to that of [Theorem 1.3](#). Here, we consider the distribution

$$P_X(n) = \begin{cases} 2^{-\lambda_\tau n} & \text{if } n \leq \tau \text{ is odd,} \\ 0 & \text{otherwise.} \end{cases}$$

It is easy to verify that  $P_X$  is indeed a valid probability distribution. Writing  $x := 2^{\lambda_\tau}$ , determining  $\lambda_\tau$  amounts to finding the roots of

$$f_\tau(x) := x^2 - x - 1 + x^{-\tau}.$$

To determine the conditions for which  $\lambda_\tau$  is the exact capacity, we proceed as in the proof of [Theorem 1.3](#) (i.e., by checking the conditions of [Lemma 2.1](#)). By routine algebraic manipulation, we conclude that this is the case if and only if

$$\frac{1}{2^{\lambda_\tau+1}} \leq p \leq \frac{1}{2}. \quad (21)$$

The convergence statement for  $C(p, \tau) = \lambda_\tau$  follows from [Lemma 6.2](#) below. Indeed, since  $0 < \varphi - 2^{\lambda_\tau} < 1.3^{-\tau}$ , it holds that

$$\log \varphi - \lambda_\tau = \log\left(\frac{\varphi}{2^{\lambda_\tau}}\right) < \frac{\varphi}{2^{\lambda_\tau}} - 1 < 1.3^{-\tau},$$

where we used that  $\log(x+1) \leq x$  for  $x > 0$ . For the limit point  $p = \frac{1}{2^{\lambda_\tau+1}}$ , we have that

$$p \leq \frac{1}{\varphi - 1.3^{-\tau} + 1} = \frac{1}{\varphi + 1} + \frac{1.3^{-\tau}}{(\varphi + 1)(\varphi + 1 - 1.3^{-\tau})} \leq \frac{1}{\varphi^2} + \frac{1.3^{-\tau}}{\varphi^2},$$

where we used the fact that  $\varphi + 1 = \varphi^2$ . □

**Lemma 6.2.** *Fix an integer  $\tau \geq 2$  and define*

$$f_\tau(x) = x^2 - x - 1 + x^{-\tau}$$

*and*

$$f(x) = x^2 - x - 1.$$

*Then, in the interval  $I = (1, 2)$ ,  $f$  has a unique root  $x^* = \varphi$  and  $f_\tau$  has a unique root  $x_\tau^*$ . Moreover,  $|x_\tau^* - x^*| < 1.3^{-\tau}$ .*

*Proof.* The claim about the root of  $f$  is easy to check. Moreover, for all  $\tau$  we have

$$f'_\tau(x) = 2x - 1 - \tau x^{-(\tau+1)}$$

and

$$f''_\tau(x) = 2 + \tau(\tau+1)x^{-(\tau+2)}.$$

Thus,  $f'_\tau(1) < 0$  and  $f'_\tau$  is strictly increasing in  $[1, 2]$ . Since  $f_\tau(1) = 0$  and  $f_\tau(\varphi) > f(\varphi) = 0$ , it must be the case that  $f_\tau$  has a unique root  $x_\tau^* \in (1, \varphi)$ . Since  $f_\tau(x) > f_{\tau+1}(x)$  for all  $x \in (1, \varphi)$ , the sequence  $x_\tau^*$  is increasing, and thus converges to some value in the interval  $[x_2^*, \varphi]$ . Note that  $f'(x) > 1$  for all  $x \in (1, 2)$ . Thus, writing  $\delta_\tau := \varphi - x_\tau^*$ , we have

$$0 = f_\tau(x_\tau^*) = f(\varphi - \delta_\tau) + (x_\tau^*)^{-\tau} < -\delta_\tau + (x_\tau^*)^{-\tau},$$

from which we conclude that  $\delta_\tau < (x_\tau^*)^{-\tau} \leq (x_2^*)^{-\tau}$ , where  $x_2^* > 1.3$ . □

**Remark 6.3** (On the optimality of naive runlength-constrained coding for even  $\tau$ ). If  $\tau$  is even then  $Y_\tau$  is not absolutely continuous with respect to  $Y$ , where  $Y$  is the output distribution corresponding to the input  $X$  defined in the proof of [Theorem 6.1](#), since  $\tau + 1 \in \text{supp}(Y_\tau) \setminus \text{supp}(Y)$ . Thus,  $D_{\text{KL}}(P_{Y_\tau} || P_Y) = +\infty$ , from which we conclude that  $X$  is not capacity-achieving by [Lemma 2.2](#).

## 6.2 Completing the region $p \in (0, \frac{1}{2}]$

We now determine the capacity  $C(p, \tau)$  of the bounded runlength channel for the region of  $p \leq \frac{1}{2}$  not covered by the previous result. Contrarily to before, we are able to determine the exact capacity for both even and odd values of  $\tau$ .

**Theorem 6.4.** Fix an integer  $\tau > 1$  and  $p \in (0, 1/2]$ , and let  $\lambda_{\tau,p} \in (0, 1)$  be the unique solution to

$$2^{\lambda p - h(p)} \cdot (1 - 2^{-\lambda(\tau+1)}) - 2^\lambda + 1 = 0. \quad (22)$$

Then,  $C(p, \tau) = \lambda_{\tau,p}$  if  $p \in (0, \frac{1}{2^{\lambda_{\tau,p}+1}}]$ . In particular,

$$C(p, \tau) > \lambda_p^* - 1.8 \cdot 1.3^{-\tau}$$

for  $p \in (0, \frac{1}{2^{\lambda_p^*+1}}(1 + 1.8 \cdot 1.3^{-\tau})]$ , where  $\lambda_p^*$  is the solution to

$$2^{\lambda p - h(p)} - 2^\lambda + 1 = 0.$$

*Proof.* The proof is similar to that of [Theorem 1.4](#). Defining

$$Q(n) = 2^{-\lambda(n-p)-h(p)}$$

for all  $n \in [\tau + 1]$ , one can easily verify that  $\sum_{n=1}^{\tau+1} Q(n) = 1$  if and only if  $\lambda \in (0, 1)$  satisfies [Equation \(22\)](#). Note that the map  $f(\lambda) = \sum_{n=1}^{\tau+1} Q(n)$  is decreasing in  $\lambda$ . Moreover,  $f(1) = 2^{p-h(p)}(1 - 2^{-(\tau+1)}) < 1$  and  $f(0) = (\tau+1)2^{-h(p)} > 1$  whenever  $p \leq \frac{1}{2}$  and  $\tau > 1$ . We then have that  $f(\lambda)$  must attain the value 1 for some unique  $\lambda_{\tau,p} \in (0, 1)$ . Equivalently, [Equation \(22\)](#) has



a unique solution  $\lambda_{\tau,p} \in (0, 1)$ . Hence,  $Q$  with this choice of  $\lambda$  gives rise to a valid probability distribution. Analogously to the proof of [Theorem 1.4](#), we can conclude that

$$D_{\text{KL}}(P_{Y_n} || Q) = \lambda n,$$

for every  $n \in [\tau]$ . We now must check that  $Q$  is a valid output distribution corresponding to some input distribution  $P_X$ . This amounts to finding some  $P_X$  supported on  $[\tau]$  such that, for all  $n \in [\tau + 1]$ ,

$$Q(n) = (1 - p)P_X(n) + pP_X(n - 1).$$

From this equation we can deduce that

$$P_X(n) = \frac{1}{1 - p} \sum_{k=0}^{n-1} Q(n - k) \left( -\frac{p}{1 - p} \right)^k \quad (23)$$

for all  $n \in [\tau]$ . It suffices to show that  $P_X(n) \geq 0$  for all  $n \in [\tau]$ , as  $\sum_{n \in [\tau]} P_X(n) = 1$  is guaranteed by construction if  $\sum_{y \in [\tau+1]} Q(y) = 1$ . Substituting  $Q(n - k) = Q(1) \cdot 2^{-\lambda_{\tau,p}(n-k-1)}$  and simplifying the geometric sum in [Equation \(23\)](#), we have that  $P_X(n) \geq 0$  if and only if

$$\left( -\frac{p 2^{\lambda_{\tau,p}}}{1 - p} \right)^n \leq 1.$$

This holds for all  $n \in [\tau]$  if and only if  $\lambda_{\tau,p} \leq \log\left(\frac{1-p}{p}\right)$ . We then conclude that  $Y$  is indeed the output distribution corresponding to a valid input distribution, and that  $\lambda_{\tau,p}$  is the exact capacity, as long as

$$0 < p \leq \frac{1}{2^{\lambda_{\tau,p}} + 1}.$$

We omit the proof of the convergence statement, as the argument is similar to that of [Theorem 6.1](#). In particular, by a similar argument to that of [Lemma 6.2](#), one can show that  $2^{\lambda_p^*} - 2^{\lambda_{\tau,p}} < 1.8 \cdot 1.3^{-\tau}$  for all values of  $p$  considered.  $\square$

We reiterate that [Theorem 6.1](#) and [Theorem 6.4](#) provide the exact capacity  $C(p, \tau)$  for all odd  $\tau$  and  $p \in (0, \frac{1}{2}]$ .

**Corollary 6.5.** *Let  $\tau > 1$  be odd and  $p \in (0, \frac{1}{2}]$ . Let  $\lambda_\tau$  and  $\lambda_{\tau,p}$  be as in [Theorems 6.1](#) and [6.4](#), respectively. Then,*

$$C(p, \tau) = \begin{cases} \lambda_\tau & \text{if } p \geq \frac{1}{2^{\lambda_\tau} + 1} \\ \lambda_{\tau,p} & \text{otherwise} \end{cases}.$$

*Proof.* The result follows from [Theorems 6.1](#) and [6.4](#) by noting that  $\lambda_\tau = \lambda_{\tau,p}$  for  $p = \frac{1}{2^{\lambda_\tau} + 1}$ .  $\square$

### 6.3 Properties of the optimal distribution for even-sized alphabet

Given that the naive runlength-constrained coding strategy is never optimal for even-sized input alphabet  $[\tau]$ , it makes sense to ask whether the capacity-achieving input distribution has full support in the region of  $p \leq \frac{1}{2}$  for which we have not determined the exact capacity. We answer this question in the negative, provided that  $\tau$  is large enough.

**Lemma 6.6.** *Let  $p \in (\frac{1}{\varphi^2}, \frac{1}{2})$ . For all  $\tau$  large enough, the capacity achieving distribution does not have full support.*

*Proof.* Fix  $p \in \left(\frac{1}{\varphi^2}, \frac{1}{2}\right)$ . For each  $\tau$ , we define  $\varepsilon_\tau$  as the quantity such that the interval of valid values of  $p$  given by [Theorem 6.4](#) is  $\left(0, \frac{1}{\varphi^2} + \varepsilon_\tau\right]$ , and we denote  $\lambda_\tau = C(p, \tau)$ .

Let  $\tau'$  be the smallest odd natural number such that  $p > \frac{1}{\varphi^2} + \varepsilon_{\tau'}$ . Suppose that  $X$  has full support and is capacity-achieving for even-sized input alphabet  $[\tau]$  (we show that this leads to a contradiction if  $\tau$  is sufficiently large). By [Lemma 2.2](#), the corresponding output distribution  $P_Y$  must satisfy

$$D(P_{Y_n} || P_Y) = \lambda_\tau n$$

for every  $n \in [\tau]$ . By the same reasoning as that of [Section 4.1](#), writing

$$P_Y(n) = 2^{-\lambda_\tau(n-p) - h(p) - g(n)}$$

for  $n \in [\tau + 1]$ , it must be the case that

$$g(n) = \left(-\frac{1-p}{p}\right)^{n-1} \beta$$

for all  $n \in [\tau + 1]$ , where  $\beta \in \mathbb{R}$  is a constant.

Since  $p > p_{\tau'}$ , we have that  $p$  belongs to the interval defined in [Theorem 6.1](#) when the input alphabet is  $[\tau']$  (i.e., naive runlength-constrained coding is optimal). By [Equation \(21\)](#), for  $\tau > \tau'$  it holds that

$$\lambda_\tau > \lambda_{\tau'} > \log\left(\frac{1-p}{p}\right). \quad (24)$$

We have seen in [Equation \(23\)](#) that, if  $P_Y$  is the output distribution induced by  $P_X$ , then

$$P_X(n) = \frac{1}{1-p} \sum_{k=0}^{n-1} P_Y(n-k) \left(-\frac{p}{1-p}\right)^k.$$

Since  $P_X$  is a probability distribution with full support,  $P_X(2) > 0$ . By the above characterization of  $P_X$ , this is equivalent to the condition

$$2^{\frac{\beta}{p} - \lambda_\tau} > \frac{p}{1-p}.$$

In particular,  $\beta$  must satisfy

$$\beta > p \cdot \left(\lambda_\tau - \log\left(\frac{1-p}{p}\right)\right) > 0. \quad (25)$$

Defining  $\delta_p := p \cdot \left(\lambda_{\tau'} - \log\left(\frac{1-p}{p}\right)\right)$ , by [Equations \(24\) and \(25\)](#) we also have that  $\beta > \delta_p > 0$ , where  $\delta_p$  is a constant not depending on  $\tau$ . On the other hand, for even  $\tau$ , a necessary condition to ensure that  $Q(\tau) \leq 1$  is that

$$\beta \leq (\lambda_\tau(\tau - p) + h(p)) \cdot \left(\frac{p}{1-p}\right)^{\tau-1}.$$

Clearly, if  $p < \frac{1}{2}$  then the right-hand side of the equation above is at most  $\delta_p$  for  $\tau$  large enough. This implies  $\beta \leq \delta_p$ , a contradiction.  $\square$

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## A Capacity of additive-noise sticky channels and capacity per unit cost of additive-noise memoryless channels

We now formalize the intuition that the capacity of additive-noise sticky channels equals the capacity of the corresponding additive-noise memoryless channel over the positive integers. We begin by recalling the framework of Verdú [Ver90], particularized to the cost function  $c(x) := x$ .

**Definition A.1** ( $((n, M, \nu, \varepsilon)$  code). *Let  $\text{Ch}$  be a channel with input alphabet  $\mathcal{A} \subseteq \mathbb{N}_{>0}$ . We say that  $\mathcal{C} \subseteq \mathcal{A}^n$  is an  $(n, M, \nu, \varepsilon)$  code with respect to  $\text{Ch}$  if it has  $|\mathcal{C}| = M$  with each codeword  $c = (c_1, \dots, c_n) \in \mathcal{C}$  satisfying*

$$\sum_{i=1}^n c_i \leq \nu,$$

*and where the average decoding error probability (taken over the uniformly random choice of  $c \in \mathcal{C}$ ) is at most  $\varepsilon$ .*

**Definition A.2** (Achievable rate per unit cost). *A rate  $R > 0$  is  $\varepsilon$ -achievable per unit cost with respect to  $\text{Ch}$  if for every  $\gamma > 0$  there exists  $\nu_0 > 0$  such that, for all  $\nu \geq \nu_0$ , there is an  $(n, M, \nu, \varepsilon)$  code with respect to  $\text{Ch}$  satisfying  $\log M > \nu(R - \gamma)$ . We say that  $R$  is achievable per unit cost with respect to  $\text{Ch}$  if it is  $\varepsilon$ -achievable per unit cost for all  $\varepsilon > 0$ .*

**Definition A.3** (Capacity per unit cost). *The capacity per unit cost of  $\text{Ch}$ , denoted by  $\text{CPUC}(\text{Ch})$ , is the supremum of the achievable rates per unit cost with respect to  $\text{Ch}$ .*

It turns out that we may assume that  $\nu$  scales linearly with the blocklength  $n$ , a fact that will be useful below.

**Lemma A.4** ([Ver90, Corollary of Theorem 2]). *A rate  $R > 0$  is achievable per unit cost with respect to Ch if and only if for every  $\varepsilon, \gamma > 0$  there exist  $c, \nu_0 > 0$  such that, for all  $\nu \geq \nu_0$ , there is an  $(n, M, \nu, \varepsilon)$  code with respect to Ch satisfying  $\log M > \nu(R - \gamma)$  and  $\nu \leq cn$ .*

### A.1 Proof of Lemma 1.2

As mentioned before, this argument is analogous to that of Mitzenmacher [Mit08, proof of Theorem 2.1]. We reproduce it here for completeness.

We begin by noting that without loss of generality we may assume that all inputs to the  $\text{ANSC}_{\mathcal{N}}$  channel begin with a 0, as this does not change the channel capacity. Under this condition, there is a straightforward one-to-one correspondence between inputs to each channel that follows by mapping an input  $x = 0^{\ell_1}1^{\ell_2}0^{\ell_3}\dots$  to the  $\text{ANSC}_{\mathcal{N}}$  to its “runlength-encoding”  $(\ell_1, \ell_2, \ell_3, \dots)$ , which will be sent through the  $\text{ANMC}_{\mathcal{N}}$ . Then, the decoding error probability of a code over the  $\text{ANSC}_{\mathcal{N}}$  equals the decoding error probability of its runlength-encoding over the  $\text{ANMC}_{\mathcal{N}}$ .

We now show that any rate  $R$  achievable per unit cost in  $\text{ANMC}_{\mathcal{N}}$  is also achievable (in the usual sense) in  $\text{ANSC}_{\mathcal{N}}$ . Let  $R$  be achievable per unit cost in  $\text{ANMC}_{\mathcal{N}}$ . By Lemma A.4, for every  $\varepsilon, \gamma > 0$ , there is a constant  $c > 0$  such that, for all  $\nu$  large enough, there exists an  $(n, M, \nu, \varepsilon)$  code  $\mathcal{C}_{\mathbb{N}}$  for  $\text{ANMC}_{\mathcal{N}}$  satisfying  $\log M > \nu(R - \gamma)$  and  $n \leq \nu \leq cn$ . By the one-to-one correspondence mentioned above,  $\mathcal{C}_{\mathbb{N}}$  is then mapped to a binary code  $\mathcal{C}$  with codeword lengths between  $n$  and  $\nu \leq cn$  and average error probability  $\varepsilon$  in  $\text{ANSC}_{\mathcal{N}}$ . We now define  $\mathcal{C}'$  to contain the codewords of  $\mathcal{C}$  with decoding error probability at most  $2\varepsilon$ . An averaging argument gives that  $M' := |\mathcal{C}'| \geq M/2$ . We further restrict ourselves to the set  $\mathcal{C}''$  of codewords of  $\mathcal{C}'$  of a fixed length  $\ell$  with  $n \leq \ell \leq \nu$ , where  $\ell$  is chosen to maximize  $\log |\mathcal{C}''|$ . We then have that

$$\frac{\log |\mathcal{C}''|}{\ell} \geq \frac{\log M'}{\ell} - \frac{\log(\nu - n + 1)}{\ell} \geq \frac{\log M}{\ell} - \frac{1 + \log(\nu - n + 1)}{\ell} > R - \gamma - o(1).$$

Thus, for  $n$  large enough, there exists a code for  $\text{ANSC}_{\mathcal{N}}$  with rate at least  $R - 2\gamma$  and maximum decoding error probability  $2\varepsilon$ , as desired.

By a similar argument, we can show that any rate  $R$  achievable in  $\text{ANSC}_{\mathcal{N}}$  is also achievable per unit cost in  $\text{ANMC}_{\mathcal{N}}$ . To this end, let  $R$  be an achievable rate in  $\text{ANSC}_{\mathcal{N}}$ . Then, for every  $\varepsilon, \gamma > 0$ , for  $n$  large enough, there exists a code  $\mathcal{C}$  for  $\text{ANSC}_{\mathcal{N}}$  of size  $M$  with blocklength  $n$ , maximum decoding error probability  $\varepsilon$ , and such that  $\log M > n(R - \gamma)$ . Consider the code  $\mathcal{C}_{\mathbb{N}}$  obtained from  $\mathcal{C}$  via runlength-encoding as described above. Then,  $\mathcal{C}_{\mathbb{N}}$  has average decoding error at most  $\varepsilon$  in  $\text{ANMC}_{\mathcal{N}}$ , constant codeword cost  $n$ , and the length of each codeword is at most  $n$ . Thus, we take  $\mathcal{C}'_{\mathbb{N}}$  to be the subset of  $\mathcal{C}_{\mathbb{N}}$  obtained by choosing the codewords of fixed length  $\ell \leq n$ , where  $\ell$  is chosen to maximize  $\log |\mathcal{C}'_{\mathbb{N}}|$ . We now have that

$$\frac{\log |\mathcal{C}'_{\mathbb{N}}|}{n} \geq \frac{\log M - \log n}{n} > R - \gamma - o(1),$$

and the result follows.